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EDUCATION

Vishwakarma Institute of Technology, Pune

Bachelor of Technology in Computer Science and Engineering - AI&ML

Nov 2024 – May 2028

CGPA: **9.33/10.0**

PROJECTS

TatvAI – Multimodal AI News Analysis Platform | Python, Flask, Whisper, Transformers, BERT, NLP

Sept 2025 – Dec 2025

- Built an **end-to-end multimodal AI pipeline** that ingests video news, extracts audio/text, and generates concise, structured summaries, accepted for presentation at the **ETLTC-ICETM 2026 International Conference (Japan)**.
- Implemented **multilingual speech-to-text transcription** using **OpenAI Whisper**, converting raw news audio into clean structured text for downstream NLP processing.
- Fine-tuned a **BERT classifier for fake-news detection** and integrated a **BART summarization model**, combining both into a unified inference pipeline for content analysis.
- Designed and shipped a **Flask REST API** and web interface supporting video upload, automated AI analysis, and real-time visualization of results.

Customer Churn Prediction using Machine Learning | Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Feb 2026

- Built a **supervised ML pipeline** to predict telecom customer churn, training and comparing **Random Forest** and **Logistic Regression** classifiers on a real-world telecom dataset.
- Engineered features and ran **EDA** with **Pandas, NumPy, Matplotlib, and Seaborn** to surface behavioral churn drivers and handle class imbalance.
- Benchmarked models using **accuracy, confusion matrix, and ROC-AUC**, selecting the best-performing classifier for deployment-ready inference.
- Ran **feature-importance analysis** to explain model predictions, identifying **contract type, tenure, and monthly charges** as the top churn signals for business stakeholders.

SkillXchange – Peer Skill-Sharing Platform | MySQL, SQL, DBMS

Oct 2025 – Nov 2025

- Designed and implemented a **normalized, persistent MySQL database** to manage users, project requests, messages, notifications, and skills with full referential integrity.
- Wrote advanced **SQL** including joins, subqueries, aggregate functions, **views, triggers, and stored procedures** to power backend business logic and reporting.
- Built analytical views and a simulated **materialized view** to efficiently summarize user activity for dashboards and reporting.

EXPERIENCE

Research Intern | IEEE CIS Pune Chapter & IEEE YP Pune Section

Jun 2026 – Jul 2026

Pune, India (Virtual)

- Built a full-stack AI-powered answer sheet evaluation system using **Node.js** and **Express.js**.
- Automated **OCR**-based handwritten evaluation, **OMR** processing, and AI-generated feedback.
- Designed **REST APIs** and integrated analytics dashboards for evaluation insights.

SKILLS

Languages: Python, C, C++, Java, SQL

Libraries / Tools: NumPy, Pandas, Matplotlib, Seaborn, Flask, Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Databases: MySQL

ML / AI: Scikit-learn, Transformers, BERT, NLP, Whisper, Model Evaluation

Data Science: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, EDA

CS Fundamentals: Data Structures, Algorithms, OOP, DBMS, Operating Systems

ACHIEVEMENTS

- * Maintained a CGPA of **9.33/10**.
- * Accepted to present two research papers at the **ETLTC-ICETM 2026 International Conference (Japan)**:
 - TatvAI** – a multimodal AI-based news analysis platform.
 - FlowWave-AI** – a real-time, cloud-native system for intelligent traffic management and dynamic green-corridor prioritization.
- * Solved **90+** Data Structures and Algorithms problems on **LeetCode**, focusing on arrays, linked lists, trees, recursion, and hashing.